

Grain Market Management System

Advanced A-to-Z Roadmap — Khata, Kisan, Fasal, Sale, Commission & Reports

Goal: Aisa practical system banana jo Grain Market ki shop ke daily kaam ko digital kare: kis se kitna lena hai, kis ko kitna dena hai, kis kisan ki kaunsi fasal aayi, kitni quantity thi, kis rate par sale hui, commission kitna bana, payment kitni hui aur kitna balance baqi hai.

1. Core System Modules

Module	Kya kaam karega
Kisan / Party Management	Kisan, buyer, supplier ya party ka profile, mobile, address aur account.
Khata / Ledger	Jama, udhaar, payment, receipt aur running balance.
Fasal Receiving	Kisan se aane wali crop ki quantity, quality, bags, weight aur date.
Sale / Trading	Buyer ko sale, rate per KG/ton, total amount aur sale details.
Commission	Percentage (%) ya per-KG/per-unit commission; automatic calculation.
Expenses / Kharchay	Loading, unloading, bardana, transport, labour, mandi fee aur other expenses.
Payments	Cash, bank, cheque/other modes; partial ya full payment.
Reports	Daily, weekly, monthly, yearly, farmer-wise, crop-wise, sale-wise aur commission reports.
Dashboard	Aaj ka business, outstanding receivable/payable, stock/crop, sales aur commission.
Backup & Security	Database backup, restore, user login aur role-based access.

2. Kisan / Party Master

Har kisan ya business party ka unique account number/ID hoga. Profile mein naam, mobile, CNIC/reference (optional), address, party type aur opening balance save hoga. Party type ko Farmer, Buyer, Supplier, Customer ya Other rakha ja sakta hai.

Important: Opening balance ko do directions mein manage karna chahiye: Lena hai (Receivable) aur Dena hai (Payable).

3. Fasal Receiving / Purchase Entry

- Date aur transaction number
- Kisan/party ka naam
- Fasal type: Gandum, Chana, Cotton, Rice, Maize, etc.
- Variety/quality/grade
- Quantity: KG, Mann, Ton ya Bags
- Gross weight, tare/empty weight aur net weight (optional advanced feature)
- Moisture / quality deductions (optional)
- Purchase/expected rate
- Bardana/packing aur transport charges
- Notes aur receipt number

4. Sale Calculation Engine

Basic formula: Total Sale = Quantity × Sale Rate.
Example: 1,000 KG × Rs. 250/KG = **Rs. 250,000**.

System mein unit conversion bhi rakhna useful hoga, taake KG, Mann aur Ton ke darmiyan calculation automatic ho. Aap apni market ke standard conversion ko settings mein define kar saken.

5. Commission Engine — 3 Options

Commission Type	Formula	Example
Percentage	Total Sale × Commission %	250,000 × 2% = Rs. 5,000
Per KG	Quantity × Commission/KG	1,000 × Rs. 5 = Rs. 5,000
Fixed	Fixed commission amount	Rs. 3,000

Recommended setting: Har transaction mein commission method select ho, lekin default method shop settings mein set kiya ja sake. Commission ko edit karne ka permission sirf authorized user ko mile.

6. Farmer Settlement / Hisab Clear

Kisan ki crop sale hone ke baad system automatically settlement statement banaye. Example: Sale Rs. 250,000 – Commission Rs. 5,000 – approved expenses Rs. 2,000 = **Farmer Net Payable Rs. 243,000**. Agar Rs. 100,000 payment ho chuki hai to **Remaining Rs. 143,000** show ho.

7. Khata / Ledger Design

Date	Description	Debit	Credit	Balance
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22-Sep	Gandum sale settlement	250,000	—	250,000
22-Sep	Commission	—	5,000	245,000
22-Sep	Cash payment	—	100,000	145,000

Ledger mein manual balance enter karne ke bajaye transactions se balance calculate ho. Is se calculation mistakes kam honggi.

8. Payments & Receipts

- Receive payment: customer/buyer se paisay receive.
- Pay payment: farmer/supplier ko paisay pay.
- Partial payment support.
- Payment method: Cash, Bank, Cheque, Other.
- Reference/cheque number aur notes.
- Printable receipt/voucher.
- Payment ko relevant ledger entry se automatically link karna.

9. Expenses & Profit Tracking

Business ki real profitability samajhne ke liye expenses module zaroori hai. Categories: transport, labour, loading, unloading, storage, commission expense, electricity, rent, misc. Har expense ko date, amount, category, party/reference aur notes ke saath record kiya ja sakta hai.

10. Reports — Daily / Monthly / Yearly

- Daily Sales Report — aaj ki total sale, quantity, average rate.
- Daily Purchase/Receiving Report — aaj kis farmer se kya crop aayi.
- Daily Commission Report — percentage/per-KG/fixed commission ka total.
- Farmer Ledger Report — selected farmer ka complete khata.
- Outstanding Receivable — kis se kitna lena hai.
- Outstanding Payable — kis ko kitna dena hai.
- Crop-wise Report — Gandum/Chana/etc. quantity aur sale.
- Party-wise Report — party ke total transactions.
- Date Range Report — From Date to To Date.
- Monthly Summary — month ki sale, purchases, commission, expenses, payments.
- Yearly Summary — saal ka overall transaction overview.
- Profit/Trading Summary — sale minus relevant costs/expenses.
- Payment Report — cash/bank/cheque wise payments.
- Stock/Quantity Report — received, sold aur remaining quantity (agar inventory maintain ki jaye).

11. Date Filter System

Reports mein preset buttons hon: Today, Yesterday, This Week, This Month, Last Month, This Year. Saath custom From Date / To Date bhi ho. Example: 01-Sep-2026 se 22-Sep-2026 tak total sale aur commission.

12. Dashboard

Card	Example
Today's Sale	Rs. 1,250,000
Today's Commission	Rs. 25,000

Card	Example
Today's Received	Rs. 500,000
Today's Paid	Rs. 300,000
Total Receivable	Rs. 4,500,000
Total Payable	Rs. 3,200,000
Pending Farmer Settlements	12

13. User Interface — Simple & Fast

Shop ke kaam mein speed important hai, is liye interface mein large buttons, quick search, keyboard shortcuts aur minimum typing rakhi jaye. Main screens: Dashboard, New Farmer, New Receiving, New Sale, Payment, Ledger, Reports, Settings.

Quick Entry: Kisan select karein → crop select karein → quantity enter karein → rate enter karein → commission method/rate → system total, commission aur net amount automatically calculate kare.

14. Database Architecture

- users — login, roles, permissions
- parties/farmers — customer/farmer master
- crops — crop master and varieties
- receipts/receiving — incoming crop transactions
- sales — crop sale transactions
- sale_items — multiple crop lines per sale
- commissions — commission rules and calculated values
- payments — incoming/outgoing payments
- expenses — business expenses
- ledger_entries — accounting movements
- stock — optional inventory balances
- settings — shop, units, commission defaults
- audit_logs — who changed what and when
- backups — backup metadata

15. Technology Roadmap

Layer	Recommended choice
Language	Python 3
Desktop UI	PySide6 (modern) or Tkinter (simple)
Database	SQLite for single-computer start; PostgreSQL/MySQL for multi-user/network setup
Reports	PDF + Excel export
Authentication	Username/password + role permissions
Backup	Automatic daily backup + manual backup/restore
Packaging	Windows executable so Python installation is not required on the shop PC

16. Security & Data Safety

- Admin, Manager aur Operator roles.
- Delete permission restricted; preferably soft-delete instead of permanent deletion.
- Every important edit/delete recorded in audit log.
- Automatic daily database backup.
- Backup copy to a second drive/cloud when available.

- Restore test every few weeks.
- Unique transaction numbers to prevent duplicate records.

17. Phase-by-Phase Development Plan

Phase 1	Requirements & database design	Final fields, units, commission rules, ledger logic.
Phase 2	Core master data	Farmers/parties, crops, users, settings.
Phase 3	Transactions	Receiving, sale, commission, payment, expenses.
Phase 4	Accounting	Ledger, receivable/payable, settlement, balances.
Phase 5	Reports	Daily/monthly/yearly/date-range and farmer/crop reports.
Phase 6	Dashboard	KPIs, pending payments, outstanding balances.
Phase 7	Security & backup	Roles, audit log, backup/restore.
Phase 8	Testing	Calculation tests, real shop scenarios, data validation.
Phase 9	Packaging	Windows app, database setup, installer/shortcut.
Phase 10	Future expansion	Multi-PC/network, barcode/weight scale integration, cloud sync, mobile access.

18. Example End-to-End Transaction

Farmer: Muhammad Ali
Crop: Gandum
Quantity: 1,000 KG
Sale Rate: Rs. 250/KG
Total Sale: $1,000 \times 250 = \text{Rs. } 250,000$
Commission: 2% = Rs. 5,000
Other approved expense: Rs. 2,000
Net farmer amount: Rs. 243,000
Payment made: Rs. 100,000
Remaining payable: Rs. 143,000

Is transaction se sale, commission, expense, payment aur farmer ledger sab linked honge. Aapko alag calculator ya register mein dobara calculation karne ki zarurat nahi hogi.

19. Advanced Features — Future Version

- Digital receipt/invoice printing.
- WhatsApp/shareable farmer statement (where supported by the deployment).
- Excel/PDF export.
- Multiple shops/branches.
- Multi-user LAN system.
- Cloud database and remote access.
- Weight scale integration for automatic weight entry.
- Barcode/QR transaction references.
- SMS/notification reminders for outstanding balances.
- Profit analytics and trend charts.
- Data import from existing Excel registers.
- Urdu/English interface toggle.

20. Final Recommended Version

Aap ke case mein first production version ko **Python + PySide6 + SQLite** par banana practical rahega. Is se ek computer par fast aur simple system banega. Jab aapko 2 ya zyada computers se same data use karna ho, database ko PostgreSQL/MySQL aur application ko LAN/network setup par move kiya ja sakta hai.

21. Build Checklist

- Farmer/Party registration
- Opening balance
- Crop master
- Receiving entry
- Sale entry
- KG/Man/Ton units
- Percentage commission

- Per-KG commission
- Fixed commission
- Automatic total calculation
- Farmer settlement
- Payments
- Expenses
- Ledger
- Receivable/Payable
- Daily report
- Monthly report
- Yearly report
- Custom date-range report
- PDF/Excel export
- Dashboard
- User roles
- Backup/restore
- Audit log
- Windows executable

Next implementation step: Is roadmap ke baad pehle database schema aur screen layout final karna chahiye, phir core Python application build karni chahiye. Sab se pehle calculation/ledger rules lock karna zaroori hai taa-ke baad mein balance aur commission mein mismatch na ho.